

# John Whitton

Principal Software Engineer | Trading Infrastructure | Execution Systems | Distributed Systems  
Rust | Market Infrastructure | DeFi Protocol Engineering  
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## Profile

Principal software engineer and technical leader specializing in Rust-based trading infrastructure, execution systems, distributed systems, and financial infrastructure.

Over the past eighteen months, I deliberately focused more deeply on hands-on engineering in trading systems, Rust, DeFi, and distributed infrastructure, designing and implementing production-oriented systems directly. This builds on more than two decades of hands-on engineering, architecture, and technical leadership across blockchain infrastructure, payments, enterprise systems, and distributed platforms.

I combine current implementation experience with system-level architecture and technical leadership, with a particular focus on performance, correctness, reliability, and operational clarity.

## Engineering Highlights

- Designed and implemented Salus, a Rust/Tokio trading and distributed-systems platform spanning real-time market ingestion, graph-based route evaluation, simulation, execution, replay, and strategy analysis.
- Built and measured a retained Ethereum topology containing 3,415 tokens, 4,552 pool components, and 2,401,108 catalog routes; retained Base workloads measured 16,537 sustained and 38,462 peak route evaluations/sec.
- Engineered bounded asynchronous processing, deterministic execution boundaries, exact arithmetic, state-freshness controls, execution telemetry, and multi-chain processing across EVM networks.
- Consulted on DeFi infrastructure spanning leverage tokens, auction-based rebalancing, Morpho lending integrations, CoW Protocol execution, ERC-6909 accounting, and protocol invariants.
- Led engineering for stablecoin payment infrastructure, cross-chain settlement, routing, proving, wallets, and Layer-2 transaction execution at Eco.
- Designed and implemented Kanga Finance, a Solidity-based DEX spanning AMMs, liquidity, staking, incentives, yield strategies, flash loans, and protocol administration.
- Designed and implemented EAVE, a Rust/Substrate Polkadot parachain and DeFi platform spanning runtime modules, staking, liquidity, EVM compatibility, APIs, and cross-chain architecture.
- Led engineering teams and technically complex initiatives, including an 8-engineer Eco team and Harmony engineering growth from 6 to 20 engineers.

## Representative Engineering Work

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>Salus (Rust)</b>             | Designed and implemented a Rust/Tokio trading platform spanning real-time market ingestion, graph construction, affected-route selection, simulation, execution, replay, and strategy analysis. Retained Ethereum topology contained 2.4M+ catalog routes; bounded asynchronous pipelines, deterministic execution boundaries, exact arithmetic, freshness controls, and execution evidence support block-sensitive multi-chain processing. | 2025–Present |
| <b>Seamless (Solidity/Rust)</b> | Solidity and Rust components for leverage tokens, auction-based rebalancing, Morpho lending integrations, CoW Protocol requests, ERC-6909 accounting, and protocol invariant analysis.                                                                                                                                                                                                                                                      | 2025–2026    |
| <b>Eco</b>                      | Stablecoin payment infrastructure, programmable money, cross-chain settlement, routing, proving, wallet services, and Layer-2 transaction execution.                                                                                                                                                                                                                                                                                        | 2023–2025    |
| <b>(Solidity/TypeScript)</b>    | One-time-password smart contract wallet eliminating seed phrase management through cryptographic OTP authentication.                                                                                                                                                                                                                                                                                                                        | 2022–2023    |
| <b>1wallet (Solidity)</b>       | ENS-style decentralized naming service for Harmony using Go and Solidity.                                                                                                                                                                                                                                                                                                                                                                   | 2022–2023    |
| <b>1ns (Go/Solidity)</b>        | Designed and implemented Solidity DEX infrastructure spanning AMM pools, multi-asset liquidity, staking, protocol incentives, yield strategies, flash loans, fee conversion, and liquidity migration.                                                                                                                                                                                                                                       | 2022–2023    |
| <b>Kanga Finance (Solidity)</b> | Designed and built a Rust/Substrate Polkadot parachain and DeFi platform spanning runtime modules, consensus integration, governance, staking, liquidity, EVM compatibility, APIs, and cross-chain architecture; deployed standalone testnet and parachain environments.                                                                                                                                                                    | 2020–2022    |
| <b>EAVE (Rust)</b>              | Layer-1 blockchain infrastructure including mainnet launch, validator onboarding, staking, SDKs, Ledger integration, APIs, and ecosystem tooling.                                                                                                                                                                                                                                                                                           | 2019–2020    |
| <b>Harmony (Go/Solidity)</b>    | Enterprise blockchain, transaction-processing APIs, tokenization, mobile applications, and partner integration platforms.                                                                                                                                                                                                                                                                                                                   | 2016–2018    |
| <b>Loyyal (Go)</b>              | Service-oriented platforms integrating SAP, Salesforce, payments, ecommerce, licensing, logistics, and partner ecosystems.                                                                                                                                                                                                                                                                                                                  | 1999–2016    |
| <b>Enterprise Integration</b>   |                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |

## Core Technologies

|                               |                                                                                                                                     |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Languages</b>              | Rust, Solidity, Go, TypeScript, Python                                                                                              |
| <b>Trading Infrastructure</b> | Market data, execution, graph-based routing, simulation, replay, strategy evaluation, MEV, DeFi                                     |
| <b>Distributed Systems</b>    | Rust, Tokio, async/concurrency, streaming, bounded queues, graph processing, deterministic execution, service-oriented architecture |
| <b>Data Platforms</b>         | PostgreSQL, ClickHouse (evaluation), MongoDB                                                                                        |
| <b>Cloud &amp; DevOps</b>     | AWS, Docker, Kubernetes, Terraform, CI/CD                                                                                           |
| <b>Observability</b>          | Datadog, PagerDuty                                                                                                                  |
| <b>Blockchain / Runtime</b>   | Ethereum/EVM, Base, Arbitrum, Unichain, Optimism, Polkadot, Substrate, FRAME, Harmony, Hyperledger                                  |
| <b>DeFi / Protocol</b>        | AMMs, liquidity pools, staking, lending, flash loans, intent execution, MEV                                                         |
| <b>Developer Tooling</b>      | Tycho, Alloy, Substreams, Foundry                                                                                                   |
| <b>AI Engineering</b>         | AI-assisted software engineering, architecture and documentation automation, code review, Codex, ChatGPT, GitHub Copilot, Claude    |

## Technical Portfolio

Selected technical evidence covering trading-system architecture, distributed systems, Rust implementation, protocol engineering, architecture decisions, and source-backed code walkthroughs. Detailed material is available for technical review.

<https://portfolio.johnwhitton.com>

## Career Experience

### Principal Engineer / Founder

2018 – Present

*Jincubator*

- Since 2025, deliberately focused more deeply on hands-on engineering, designing and implementing Rust trading infrastructure, DeFi protocols, distributed systems, and supporting technical research.
- Designed and implemented Salus across market ingestion, graph-based evaluation, simulation, execution, replay, strategy analysis, telemetry, and deterministic evidence boundaries.
- Consulted on DeFi leverage-token infrastructure, auction-based rebalancing, lending integrations, intent execution, and protocol invariants.
- Built and contributed to blockchain and protocol systems including 1wallet, 1ns, Kanga Finance, EAVE, solver/intent infrastructure, and supporting engineering tooling.
- Applied AI-assisted engineering workflows across implementation, architecture, testing, documentation, and code review while retaining deterministic engineering and validation boundaries.

### Engineering Manager / Senior Protocol Engineer

Aug 2023 – Jan 2025

*Eco Inc*

- Led a team of 8 engineers building stablecoin payment infrastructure, programmable money systems, routing, proving, wallet, and protocol services.
- Delivered cross-chain routing, settlement, and transaction execution infrastructure across Ethereum and Layer-2 ecosystems while recruiting engineers and managing delivery.

### Ecosystem Architect

Aug 2019 – May 2020

*Harmony Protocol*

- Supported Harmony Mainnet launch, staking infrastructure, APIs, SDKs, validator tooling, and ecosystem integrations.
- Coordinated 50+ production network upgrades, deployed 300+ nodes, supported 600+ validators, and scaled engineering from 6 to 20 engineers.

### Chief Technology Officer

Jul 2016 – Jun 2018

*Loyyal*

- Led distributed teams building enterprise blockchain, loyalty, transaction-processing APIs, Hyperledger, private Ethereum, and partner integration platforms.

### Earlier Experience

*Enterprise Architect / Engineering Manager / Founder — EFI, Autodesk, Global Enterprise Solutions*

- Built enterprise SOA platforms spanning SAP, Salesforce, licensing, ecommerce, payments, logistics, and partner systems.
- Led architecture across 100+ enterprise initiatives, managed \$3M+ planning portfolios, and founded a consulting firm generating approximately \$800K annual revenue.

## Education

**Bachelor of Computer Science**, University of Queensland